

Pre-label Tool User Guide

What is this tool?

This is a multi-view pre-labeling tool for Orin recordings. It lets a user view recorded multi-view sequences in a browser, fill metadata, draw bounding boxes, submit labels, upload completed sequences to PDX/CSS, and move to the next ready sequence.

Main Orin recordings path:

```
/mnt/nova_ssd/recordings
```

Main workflow

```
Orin recordings
/mnt/nova_ssd/recordings/<sequence_name>/
  → Pre-label UI reads local videos from Orin
  → If MP4s are missing, app extracts them from MCAP automatically
  → User fills metadata and draws bounding boxes
  → Submit uploads videos + hoi_metadata.yaml to PDX/CSS
  → Local completion markers are written
  → Completed sequence disappears
  → Next ready sequence appears
```

Important setup note

Setup is needed only once on the shared Orin. If the repo already exists at `/home/nvidia/prelabel-tool`, users do not need to clone or install again. Most users only need the Daily Usage steps.

Shared Orin:

```
nvidia@nova-devkit-6.dyn.nvidia.com
```

Required sequence format

Each sequence is expected under:

```
/mnt/nova_ssd/recordings/<sequence_name>/
```

The UI uses these 4 MP4 files:

```
front_stereo_camera_left.mp4
back_stereo_camera_left.mp4
left_stereo_camera_left.mp4
right_stereo_camera_left.mp4
```

If these MP4s are missing, the app automatically creates them from the sequence MCAP when `AUTO_EXTRACT_MCAP=1` is set.

Daily Usage: Run from Orin

Terminal 1: start the app on Orin

Open a terminal on your Mac and SSH into Orin:

```
ssh nvidia@nova-devkit-6.dyn.nvidia.com
```

Go to the repo:

```
cd /home/nvidia/prelabel-tool
```

Activate the Python environment:

```
source venv/bin/activate
```

Set credentials:

```
export CSS_ACCESS_KEY=YOUR_ACCESS_KEY
export CSS_SECRET_KEY=YOUR_SECRET_KEY
export AWS_ACCESS_KEY_ID="$CSS_ACCESS_KEY"
export AWS_SECRET_ACCESS_KEY="$CSS_SECRET_KEY"
export AWS_DEFAULT_REGION="us-east-1"
```

Start the app:

```
CONFIG_PATH=config_prod.yaml ALLOW_PROD_UPLOAD=1 LOCAL_RECORDINGS_DIR=/mnt/nova_ssd/recordings ONE_AT_A_TIME=1
AUTO_OPEN_BROWSER=0 AUTO_EXTRACT_MCAP=1 python app_orin.py
```

Leave this terminal open. Expected output includes:

```
Running on http://127.0.0.1:8000
```

Notes:

`AUTO_OPEN_BROWSER=0` means the app will not try to open a browser on Orin.

`AUTO_EXTRACT_MCAP=1` means the app will automatically create the four required MP4 videos from the sequence MCAP if they are missing.

If MP4s already exist, the app uses them directly. The first time a sequence is shown, it may take extra time while videos are extracted. After that, the MP4s remain in the sequence folder and load faster.

How to open another terminal

You need a second terminal for the SSH tunnel.

- **Mac Terminal:** `Cmd + N` for new window, or `Cmd + T` for new tab.
- **VS Code:** use `Terminal` → `New Terminal`.

Terminal 2: open SSH tunnel from Mac

```
ssh -N -L 8001:127.0.0.1:8000 nvidia@nova-devkit-6.dyn.nvidia.com
```

This terminal may look like it is stuck. That is normal. Leave it open.

Open the browser

```
http://127.0.0.1:8001
```

You should see one ready sequence.

How to pre-label a sequence

1. Open the displayed sequence.
2. Fill required metadata: `calib_sequence`, `object_id`, `person_id`, `action_description`, `action_language`.
3. Pause the video where the object is clearly visible.
4. Draw one bounding box for each camera view: front, back, left, and right.
5. Click **Submit**.

After submit, the completed sequence uploads to PDX/CSS and the next ready sequence appears.

After Submit, Relabeling, and Cleanup

What happens after submit?

The tool uploads these files to PDX/CSS:

```
front_stereo_camera_left.mp4
back_stereo_camera_left.mp4
left_stereo_camera_left.mp4
right_stereo_camera_left.mp4
hoi_metadata.yaml
```

It also writes local completion files on Orin:

```
/mnt/nova_ssd/recordings/<sequence_name>/hoi_metadata.yaml
/mnt/nova_ssd/recordings/<sequence_name>/prelabel_uploaded
```

These local files make the sequence disappear from the ready list.

Relabel a sequence

If the label was wrong and the sequence needs to be labeled again, delete only the local completion files:

```
rm /mnt/nova_ssd/recordings/<sequence_name>/hoi_metadata.yaml
rm /mnt/nova_ssd/recordings/<sequence_name>/prelabel_uploaded
```

Then run the app again. The sequence will show again. Submitting again uploads to the same PDX/CSS path. Deleting local completion files does not delete anything from PDX/CSS.

Free Orin disk space

After confirming the upload is complete, delete the local Orin sequence folder to free space:

```
rm -rf /mnt/nova_ssd/recordings/<sequence_name>/
```

This deletes only local Orin files. It does not delete the uploaded sequence from PDX/CSS.

Bulk cleanup in small batches

Preview 5 completed folders:

```
cd /home/nvidia/prelabel-tool && source venv/bin/activate && CLEANUP_LIMIT=5
LOCAL_RECORDINGS_DIR=/mnt/nova_ssd/recordings python cleanup_completed_orin_sequences.py
```

Actually delete 5 completed local folders:

```
cd /home/nvidia/prelabel-tool && source venv/bin/activate && DRY_RUN=0 ALLOW_DELETE_LOCAL=1 CLEANUP_LIMIT=5
LOCAL_RECORDINGS_DIR=/mnt/nova_ssd/recordings python cleanup_completed_orin_sequences.py
```

This only deletes local Orin folders that have `.prelabel_uploaded`. It does not delete anything from PDX/CSS.

Run one specific sequence

Use this for testing one known sequence:

```
cd /home/nvidia/prelabel-tool
source venv/bin/activate
CONFIG_PATH=config_prod.yaml ALLOW_PROD_UPLOAD=1 LOCAL_RECORDINGS_DIR=/mnt/nova_ssd/recordings ONE_AT_A_TIME=1
AUTO_OPEN_BROWSER=0 AUTO_EXTRACT_MCAP=1 SEQUENCE_NAME=<sequence_name> python app_orin.py
```

Optional: run with local downloaded recordings

Main usage is Orin. This is only if recordings are already downloaded locally.

Example local path:

```
/Users/<your_user>/recordings
```

Run locally:

```
cd /path/to/prelabel-tool
source venv/bin/activate
export CSS_ACCESS_KEY=YOUR_ACCESS_KEY
export CSS_SECRET_KEY=YOUR_SECRET_KEY
export AWS_ACCESS_KEY_ID="$CSS_ACCESS_KEY"
export AWS_SECRET_ACCESS_KEY="$CSS_SECRET_KEY"
export AWS_DEFAULT_REGION="us-east-1"
CONFIG_PATH=config_prod.yaml ALLOW_PROD_UPLOAD=1 LOCAL_RECORDINGS_DIR=/Users/<your_user>/recordings ONE_AT_A_TIME=1
AUTO_EXTRACT_MCAP=1 python app_orin.py
```

Open directly:

```
http://127.0.0.1:8000
```

No SSH tunnel is needed for local mode.

One-time Setup and Troubleshooting

One-time setup on Orin

Only needed once on the shared Orin.

```
ssh nvidia@nova-devkit-6.dyn.nvidia.com
cd /home/nvidia
git clone https://github.com/rishitat-tech/prelabel-tool.git
cd /home/nvidia/prelabel-tool
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

After this, users should follow Daily Usage.

Troubleshooting

python cannot open app_orin.py

This means you are in the wrong folder.

```
cd /home/nvidia/prelabel-tool
```

Browser does not open

Check app on Orin:

```
curl -I http://127.0.0.1:8000
```

Expected:

```
HTTP/1.1 200 OK
```

Check tunnel on Mac:

```
curl -I http://127.0.0.1:8001
```

Expected:

```
HTTP/1.1 200 OK
```

No sequences show up

Check the sequence folder exists:

```
ls -lah /mnt/nova_ssd/recordings/<sequence_name>/
```

Check if it is already marked complete:

```
ls -lah /mnt/nova_ssd/recordings/<sequence_name> | grep -E "hoi_metadata|prelabel"
```

If MP4s are missing but MCAP exists, make sure the app command includes `AUTO_EXTRACT_MCAP=1`.

Quick Command Summary

Terminal 1

```
ssh nvidia@nova-devkit-6.dyn.nvidia.com
cd /home/nvidia/prelabel-tool
source venv/bin/activate
export CSS_ACCESS_KEY=YOUR_ACCESS_KEY
export CSS_SECRET_KEY=YOUR_SECRET_KEY
export AWS_ACCESS_KEY_ID="$CSS_ACCESS_KEY"
export AWS_SECRET_ACCESS_KEY="$CSS_SECRET_KEY"
export AWS_DEFAULT_REGION="us-east-1"
CONFIG_PATH=config_prod.yaml ALLOW_PROD_UPLOAD=1 LOCAL_RECORDINGS_DIR=/mnt/nova_ssd/recordings ONE_AT_A_TIME=1
AUTO_OPEN_BROWSER=0 AUTO_EXTRACT_MCAP=1 python app_orin.py
```

Terminal 2

```
ssh -N -L 8001:127.0.0.1:8000 nvidia@nova-devkit-6.dyn.nvidia.com
```

Browser

```
http://127.0.0.1:8001
```